



NORTH SYDNEY BOYS' HIGH SCHOOL
2008 HSC Course Assessment Task 3

MATHEMATICS (ADVANCED)

General instructions

- Working time – 60 minutes.
- Write in the booklet provided.
- Each new question is to be started on a new booklet.
- Write using blue or black pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets within this paper and hand to examination supervisors.
- Unless otherwise specified, $\log$ means logarithm to base e .

Class teacher (please ✓)

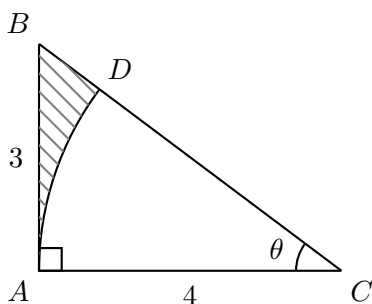
- ☐ Mr Lam
- ☐ Mr Taylor
- ☐ Mr Fletcher
- ☐ Mr Lowe
- ☐ Mr Ireland

STUDENT NUMBER: **PAGES USED:**

Marker's use only.

PART	A	B	C	D	E	Total	%
MARKS	$\overline{15}$	$\overline{12}$	$\overline{14}$	$\overline{10}$	$\overline{11}$	$\overline{62}$	

- | Part A (15 Marks) | Commence a new sheet. | Marks |
|--------------------------|--|--------------|
| 1 | Write correct to 3 decimal places: | |
| | (a) 45° to radians. | 2 |
| | (b) $\cos 2.9$. | 2 |
| 2 | What is the period and amplitude of the curve $y = 3 \cos 2x$?
Sketch the curve for $0 \leq x \leq 2\pi$. | 4 |
| 3 | From a right angled triangle $\triangle ABC$ with the right angle at A , an arc is drawn from A with centre C and radius $AC = 4$ cm. This arc meets the hypotenuse at D . This is shown in the diagram below. | |



- | | | |
|-----|---|----------|
| (a) | Find $\angle ACB$ in radians correct to 2 decimal places. | 2 |
| (b) | Find the length of the arc AD correct to 2 decimal places. | 2 |
| (c) | Find the area bounded by the region AB , BD and arc AD correct to 2 decimal places. | 3 |

- | Part B (12 Marks) | Commence a new sheet. | |
|--------------------------|------------------------------|--|
|--------------------------|------------------------------|--|

- | | | |
|----------|---|----------|
| 1 | Differentiate: | |
| | (a) $\frac{\tan x}{2x+1}$. | 2 |
| | (b) $\sin^3 x$. | 2 |
| | (c) $x^3 e^{-3x}$. | 2 |
| | (d) $\log_e \left(\frac{2x+1}{3x-7} \right)$. | 3 |
| 2 | Find the equation of the normal to the curve $y = x \sin x$ where $x = \frac{\pi}{2}$. | 3 |

Part C (14 Marks)		Commence a new sheet.	Marks
1	Sketch $y = \log_{10}(x - 2)$, showing essential features. State its domain and range.		3
2	Find the primitive of:		
(a)	$\frac{2x}{x^2 + 1}$.		2
(b)	$\frac{e^{2x}}{e^{2x} + 4}$.		2
(c)	$\frac{2}{x} + 5e^x$.		2
3	The minute hand of a town clock is 1.75 m long.		
(a)	How far does the tip move in 35 min? (Answer correct to 1 d.p.)		3
(b)	How long does it take the hand to rotate through $\frac{\pi}{15}$ radians?		2

Part D (10 Marks)		Commence a new sheet.	
1	Find the <i>exact</i> area of a minor segment of a circle, radius $5\sqrt{2}$ cm, cut off by a 10 cm chord.		3
2	(a) Sketch the curve $f(x) = 2e^{-x}$, clearly showing the y intercept. Using this, draw $y = -f(x)$.		2
	(b) $y = f(-x)$.		1
3	Evaluate:		
(a)	$\int_{\frac{\pi}{2}}^{\pi} \cos 2x \, dx$		2
(b)	$\int_0^{\frac{\pi}{4}} \frac{1}{2}x - \sin(2x) \, dx$.		2

Part E (11 Marks)		Commence a new sheet.	
1	(a) Show that $\frac{d}{dx}(x \ln x - x) = \ln x$.		2
	(b) Hence or otherwise, find		2
	$\int \ln(x^2) \, dx$		
2	(a) What are the coordinates and the nature of the stationary point for the curve $y = \frac{e^x}{x^2 + 1}$?		4
	(b) What is the range of this function? Sketch this curve, showing any intercepts.		3

End of paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; \quad x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right) + C$$

NOTE: $\ln x = \log_e x$, $x > 0$

Solutions

Part B

Part A

1 (a) (2 marks)

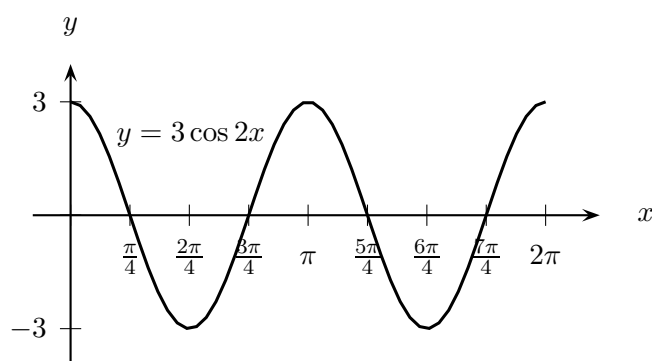
$$\frac{43^\circ \times \pi}{180^\circ} = 0.750 \text{ (3 d.p.)}$$

(b) (2 marks)

$$\cos 2.9 = -0.971 \text{ (3 d.p.)}$$

2 (4 marks)

$$\bullet \quad T = \frac{2\pi}{2} = \pi \quad \bullet \quad a = 3$$



(b) (2 marks)

$$y = \sin^3 x = (\sin x)^3$$

$$y(u) = u^3 \quad u(x) = \sin x$$

$$y'(u) = 3u^2 \quad u'(x) = \cos x$$

$$y'(x) = y'(u) \times u'(x) \\ = 3 \sin^2 x \cos x$$

(c) (2 marks)

$$y = x^3 e^{-3x}$$

$$u = x^3 \quad v = e^{-3x}$$

$$u' = 3x^2 \quad v' = -3e^{-3x}$$

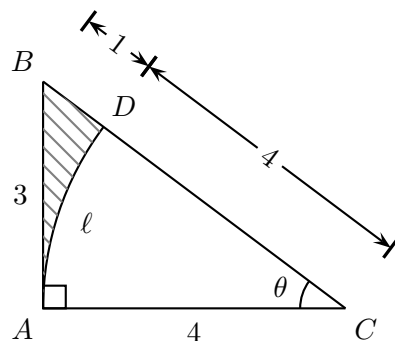
$$\frac{dy}{dx} = uv' + vu' \\ = -3x^3 e^{-3x} + 3x^2 e^{-3x} \\ = 3x^2 e^{-3x} (1 - x)$$

(d) (3 marks)

$$y = \log_e \left(\frac{2x+1}{3x-7} \right) \\ = \log_e(2x+1) - \log_e(3x-7)$$

$$\frac{dy}{dx} = \frac{2}{2x+1} - \frac{3}{3x-7} \\ = \frac{-17}{(2x+1)(3x-7)}$$

3 (a) (2 marks)



$$\tan \theta = \frac{3}{4} \\ \theta = \tan^{-1} \frac{3}{4} = 0.64$$

(b) (2 marks)

$$\ell = r\theta = 4 \times \tan^{-1} \frac{3}{4} = 2.57 \\ = 2.56 \text{ (if using 0.64 as } \theta \text{)}$$

(c) (3 marks)

$$A_{ABD} = A_{\triangle} - A_{\text{sect}} \\ = \left(\frac{1}{2}bh \right) - \left(\frac{1}{2}r^2\theta \right) \\ = \left(\frac{1}{2} \times 3 \times 4 \right) - \left(\frac{1}{2} \times 4^2 \times \tan^{-1} \frac{3}{4} \right) \\ = 6 - 5.148 \dots = 0.852$$

2 (3 marks)

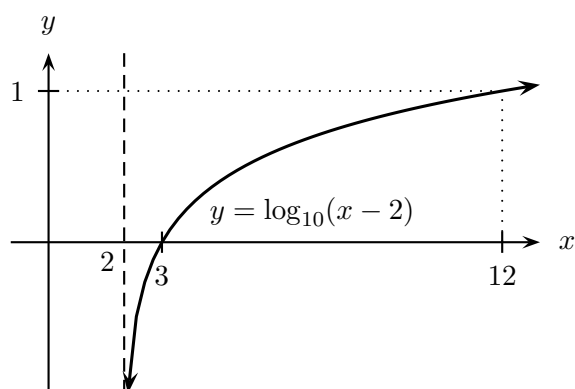
$$\begin{aligned}
 y &= x \sin x \\
 u &= x \quad v = \sin x \\
 u' &= 1 \quad v' = \cos x
 \end{aligned}$$

$$\begin{aligned}
 y' &= x \cos x + \sin x \Big|_{x=\frac{\pi}{2}} \\
 &= \frac{\pi}{2} \cos\left(\frac{\pi}{2}\right) + \sin\left(\frac{\pi}{2}\right) = 1 \\
 \therefore m_{\perp} &= -1
 \end{aligned}$$

$$\begin{aligned}
 y &= x \sin x \Big|_{x=\frac{\pi}{2}} \\
 &= \frac{\pi}{2} \sin\left(\frac{\pi}{2}\right) = \frac{\pi}{2}
 \end{aligned}$$

At $\left(\frac{\pi}{2}, \frac{\pi}{2}\right)$, $m_{\perp} = -1$. Using the point-gradient formula,

$$\begin{aligned}
 \frac{y - \frac{\pi}{2}}{x - \frac{\pi}{2}} &= -1 \\
 y - \frac{\pi}{2} &= \frac{\pi}{2} - x \\
 +\frac{\pi}{2} \quad +\frac{\pi}{2} \\
 y &= \pi - x
 \end{aligned}$$

Part C**1** (3 marks)

$$D = \{x : x \in \mathbb{R}, x > 2\} \quad R = \{y : y \in \mathbb{R}\}$$

2 (a) (2 marks)

$$\int \frac{2x}{x^2 + 1} dx = \log_e(x^2 + 1) + C$$

(b) (2 marks)

$$\begin{aligned}
 \int \frac{e^{2x}}{e^{2x} + 4} dx &= \frac{1}{2} \int \frac{2e^{2x}}{e^{2x} + 4} dx \\
 &= \frac{1}{2} \log_e(e^{2x} + 4) + C
 \end{aligned}$$

(c) (2 marks)

$$\int \frac{2}{x} + 5e^x dx = 2 \log_e(x) + 5e^x + C$$

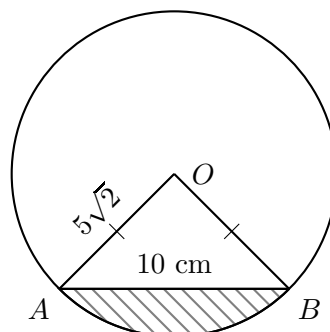
3 (a) (3 marks)

- 1 hr $\rightarrow 2\pi$
- 1 min $\rightarrow \frac{\pi}{30}$
- 35 min $\rightarrow 35 \times \frac{\pi}{30} = \frac{7\pi}{6}$

$$\ell = r\theta = 1.75 \times \frac{7\pi}{6} = 6.4 \text{ m}$$

(b) (2 marks)

Given $\frac{\pi}{30} \rightarrow 1 \text{ min}$, then $\frac{\pi}{15} \rightarrow 2 \text{ min}$.

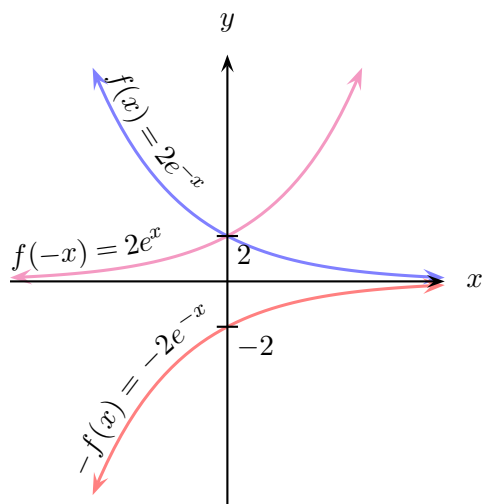
Part D**1** (3 marks)

$$\begin{aligned}
 \cos \angle AOB &= \frac{(5\sqrt{2})^2 + (5\sqrt{2})^2 - 10^2}{2 \times (5\sqrt{2}) (5\sqrt{2})} \\
 &= \frac{50 + 50 - 100}{2 \times (5\sqrt{2}) (5\sqrt{2})} = 0 \\
 \therefore \angle AOB &= \frac{\pi}{2}
 \end{aligned}$$

$$\begin{aligned}
 A &= \frac{1}{2} r^2 (\theta - \sin \theta) \\
 &= \frac{1}{2} (5\sqrt{2})^2 \left(\frac{\pi}{2} - \sin \frac{\pi}{2} \right) \\
 &= \frac{1}{2} \times 50 \left(\frac{\pi}{2} - 1 \right) = 25 \left(\frac{\pi}{2} - 1 \right) \text{ cm}^2
 \end{aligned}$$

- 2 (a) (2 marks) See next diagram – [1] for each correct curve.

- (b) (1 mark)



- (b) (2 marks)

$$\begin{aligned}\int \ln(x^2) dx &= 2 \int \ln x dx \\ &= 2(x \ln x - x) + C \\ &= 2x \ln x - 2x + C\end{aligned}$$

- 2 (a) (4 marks)

$$\begin{aligned}y &= \frac{e^x}{x^2 + 1} \\ u &= e^x \quad v = x^2 + 1 \\ u' &= e^x \quad v' = 2x \\ \frac{dy}{dx} &= \frac{vu' - uv'}{v^2} \\ &= \frac{e^x(x^2 + 1) - 2xe^x}{(x^2 + 1)^2} \\ &= \frac{e^x(x^2 - 2x + 1)}{(x^2 + 1)^2} \\ &= \frac{e^x(x - 1)^2}{(x^2 + 1)^2}\end{aligned}$$

- 3 (a) (2 marks)

$$\begin{aligned}\int_{\pi/2}^{\pi} \cos 2x dx &= \frac{1}{2} \sin 2x \Big|_{\pi/2}^{\pi} \\ &= \frac{1}{2} \left(\sin \left(2 \cdot \frac{\pi}{2} \right) - \sin(2 \cdot \pi) \right) \\ &= 0\end{aligned}$$

- (b) (2 marks)

$$\begin{aligned}&\int_0^{\pi/4} \frac{1}{2} x - \sin 2x dx \\ &= \frac{1}{4} x^2 + \frac{1}{2} \cos 2x \Big|_0^{\pi/4} \\ &= \frac{1}{4} \left(\left(\frac{\pi}{4} \right)^2 - 0 \right) \\ &\quad + \frac{1}{2} \left(\cos \left(2 \cdot \frac{\pi}{4} \right) - \cos 0 \right) \\ &= \frac{\pi^2}{64} - \frac{1}{2}\end{aligned}$$

x	$\frac{1}{2}$	1	$\frac{3}{2}$
$\frac{dy}{dx}$	+	0	+
y	$\frac{4e^{1/2}}{5}$	$\frac{e}{2}$	$\frac{2e^{3/2}}{5}$

- Horizontal point of inflexion at $(1, \frac{e}{2})$.

- (b) (3 marks)

$$y = \frac{e^x}{x^2 + 1}$$

Since $e^x > 0$ and $x^2 + 1 > 0 \forall x \in \mathbb{R}$, then

$$R = \{y : y \in \mathbb{R}, y > 0\}$$

Part E

- 1 (a) (2 marks)

$$\begin{aligned}y &= x \ln x - x \\ u &= x \quad v = \ln x \\ u' &= 1 \quad v' = \frac{1}{x} \\ \frac{dy}{dx} &= x \cdot \frac{1}{x} + \ln x - 1 \\ &= \ln x\end{aligned}$$

